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# California Housing Price Prediction Using Linear Regression

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Machine Learning Assignment Summary

EDA | Model | Performance | Insights

# Linear Regression Model

## Dataset Overview

**Dataset:** California Housing Dataset (sklearn.datasets)

**Target Variable:** Median House Value (MedHouseVal)

**Features (8):**

MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude

**Samples:** 20,640 California census block groups

**Preprocessing:**

- StandardScaler for feature normalization
- Train-test split: 80% / 20%

**No major missing-value issues detected during EDA.**

## Model & Training

**Algorithm:** Ordinary Least Squares Linear Regression (scikit-learn)

**Objective:** Minimize the sum of squared residuals between predicted and actual house values

**Training Process:**

1. Load and explore the dataset
2. Scale features using StandardScaler
3. Split into training (80%) and test (20%) sets
4. Fit Linear Regression model on training data
5. Generate predictions on the test set
6. Evaluate using MAE, MSE, RMSE, and  $R^2$

**Assumption:** Linear relationship between features and target variable

# Exploratory Data Analysis (EDA)

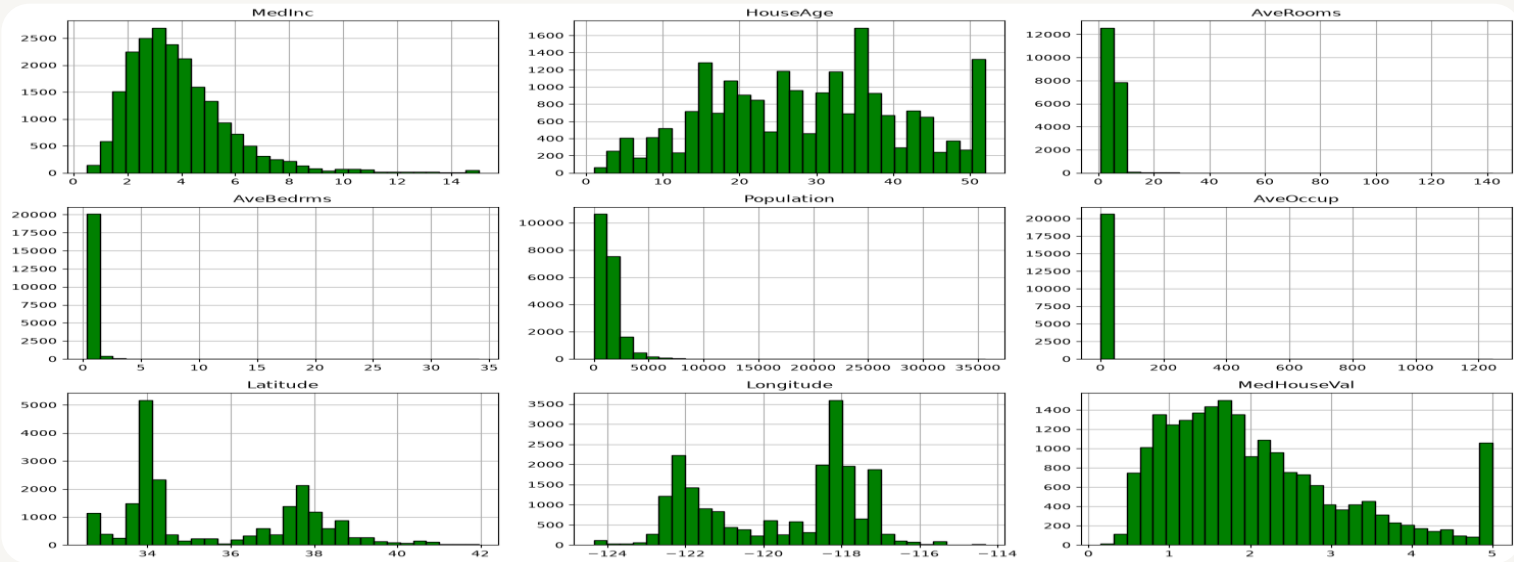


Fig.1 – Feature Histograms

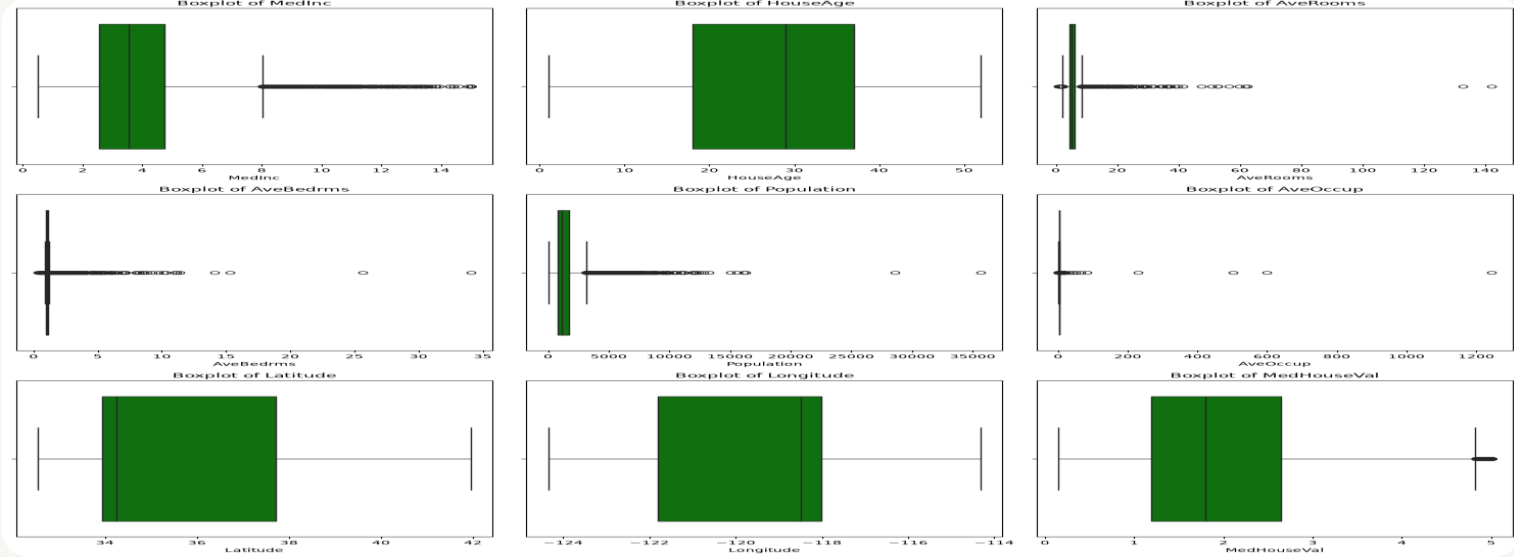


Fig.1 – Feature Boxplots

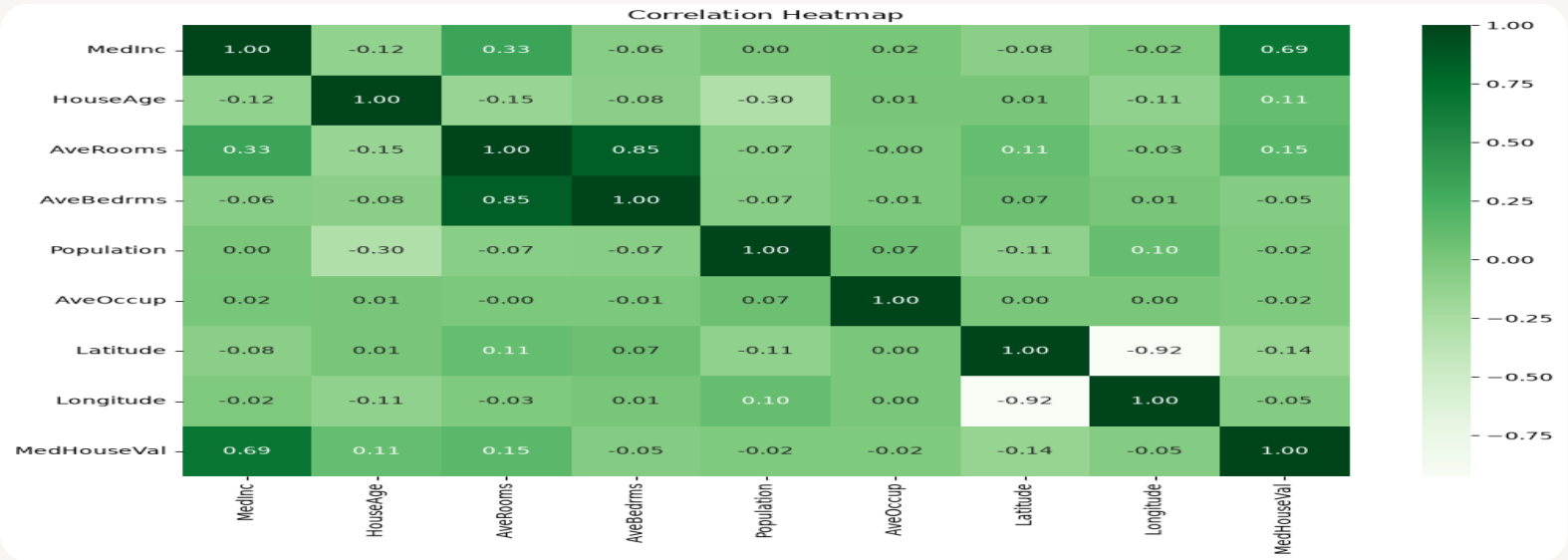


Fig.2 – Correlation Heatmap

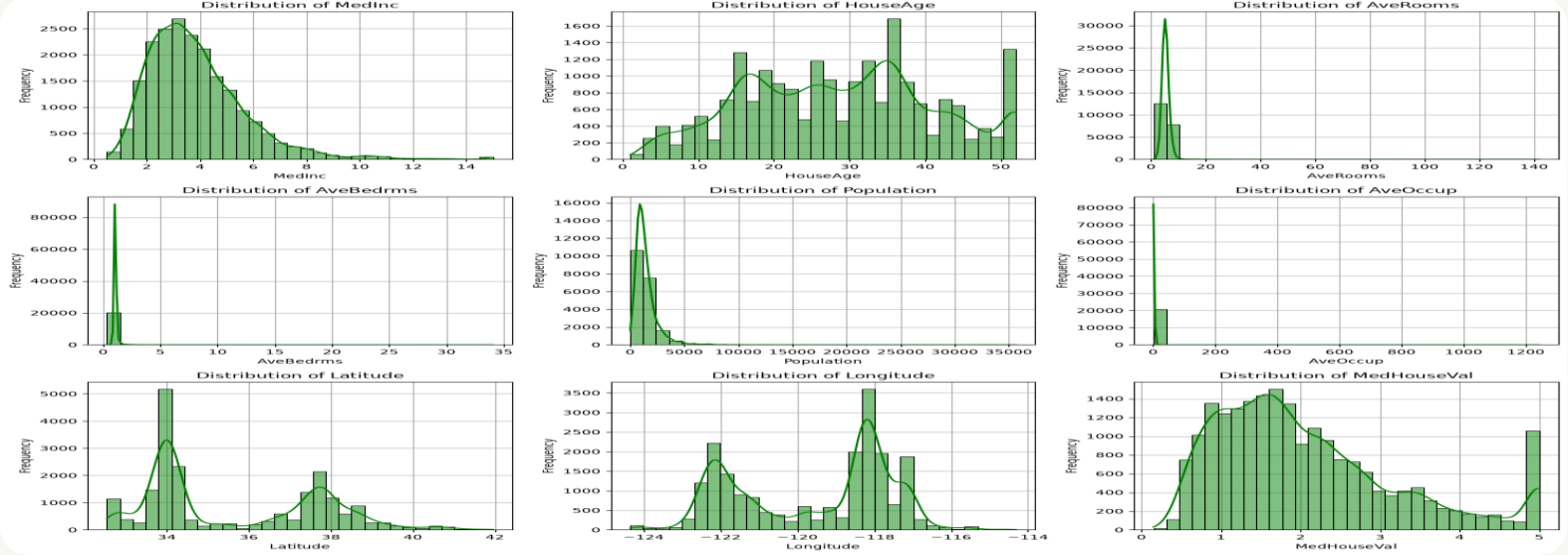


Fig.3 – Distribution Plots with KDE

## Key Observations

**MedInc** shows right skew; **HouseAge** is multimodal. **AveRooms**, **AveBedrms**, **Population**, **AveOccup** are heavily right-skewed with significant outliers. Correlation heatmap: **MedInc** is the strongest predictor of **MedHouseVal** ( $r = 0.69$ ). Latitude and Longitude correlate at  $-0.92$ , confirming geographic clustering. No major missing-value issues observed.

# Model Performance

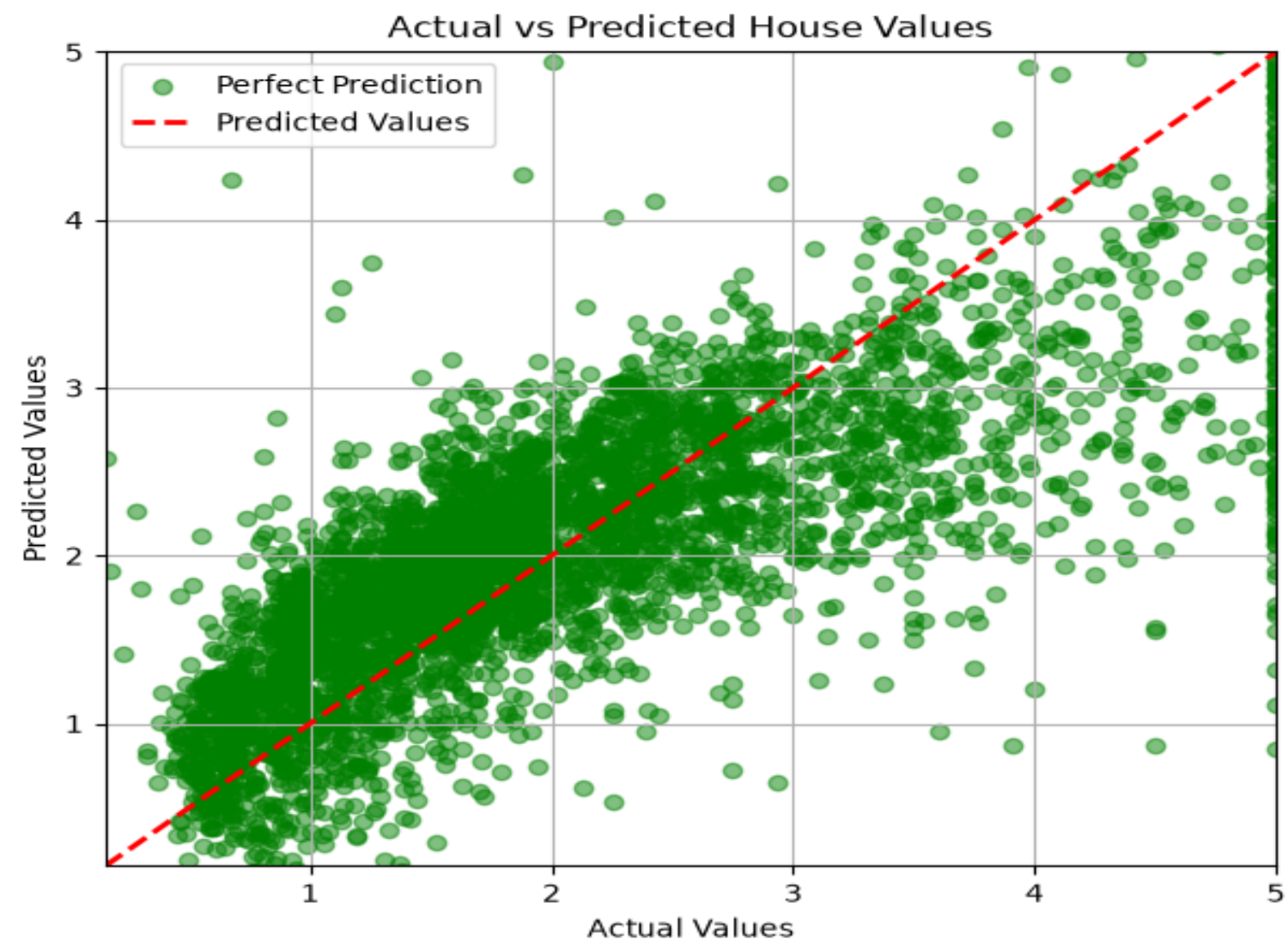


Fig.4 — Actual vs. Predicted House Values

The scatter plot shows moderate alignment between actual and predicted values. Points cluster around the ideal prediction line, but significant spread indicates room for improvement.

## Performance Metrics

| Metric                         | Value  |
|--------------------------------|--------|
| Mean Absolute Error (MAE)      | 0.5332 |
| Mean Squared Error (MSE)       | 0.5559 |
| Root Mean Squared Error (RMSE) | 0.7456 |
| R-squared (R <sup>2</sup> )    | 0.5758 |

## Performance Summary

**MAE = 0.5332** — relatively low average prediction error.

**RMSE = 0.7456** — moderate prediction deviations; larger errors penalized more heavily.

**R<sup>2</sup> = 0.5758** — the model explains **57.6%** of the variance in housing prices.

The model provides a **reasonable baseline** for predicting California house prices, with clear opportunities for improvement through advanced techniques.

# Improvement Ideas & Conclusion

## Improvement Ideas

### Feature Engineering

Create polynomial features and interaction terms to capture non-linear relationships between variables.

### Polynomial Regression

Extend the linear model with squared and higher-order terms to fit curved patterns in the data.

### Ensemble Methods

Random Forest, Gradient Boosting, and XGBoost can model complex feature interactions more effectively.

### Hyperparameter Tuning

Grid search or random search with K-fold cross-validation for robust, optimized model evaluation.

## Conclusion

Successfully implemented a **Linear Regression** model for California housing price prediction.

Conducted comprehensive **EDA** to understand feature distributions, correlations, and data quality.

Evaluated the model using **MAE, MSE, RMSE, and  $R^2$**  metrics.

The model achieved  **$R^2 = 0.5758$** , indicating moderate predictive performance.

More advanced regression models and additional feature engineering can further improve accuracy.

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